

Blockchain-Based Transparent and Secure Academic Grading Using Decentralized Verification

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Abstract. Academic integrity and reliable assessment of academic achievements play a vital role in higher education. Academic result management systems typically rely on centralized database and manual verification, and carry the risk of unauthorized data changes, falsified results, and delayed verification by third parties. This paper presents the concept of a four-component framework for secure management of academic results without compromising privacy. The system splits the academic result life cycle into distinct parts: controlled result ingestion and provenance (Component 3); academic governance including Board of Examiners (BOE) review, version tracking and finalization (Component 2); cryptographic integrity anchoring with Merkle Trees, IPFS storage and smart contracts (Component 1); and independent claim verification with Zero-Knowledge Proofs (Component 4). The deployment of the cloud-based research prototype showcases the successful end-to-end integration of the four components: a Hardhat EVM environment, a MongoDB Atlas deployment, and a Pinata IPFS deployment. It demonstrates how to achieve data integrity throughout the blockchain, IPFS dataset, record hashes, and proof index without revealing the raw grades. This work offers a scalable and decentralized blueprint for transforming academic credentialing by moving away from the monolithic reliance on blockchain and towards a trustful chain that can include data provenance, academic governance structures, and privacy-preserving verification, thereby overcoming the limitations of third-party verification.

Keywords: Academic Result Verification, Blockchain, Merkle Tree, SHA-256, Zero-Knowledge Proof, Academic Governance, Data Integrity, Decentralized Verification.

1 INTRODUCTION

Academic results are important for graduates when applying for employment, higher education, and professional opportunities. However, conventional academic result systems often rely on centralized databases and institution-controlled verification procedures. Third parties may thus have to look to issuing institutions to verify authenticity of academic claims or rely on central systems for verification.

The key challenge is not limited to protecting stored academic records from unauthorized modification; it also involves ensuring that the academic review and decision-making process leading to a finalized result is controlled, traceable, and verifiable. An academic grade passes through multiple stages before being finalized. Submissions by lectures may have to be reviewed, corrected, moderated and have special concerns that have to be approved before they are finalized. Thus, there should be a secure verification system that not only ensures the result is secure, but also the verification process that determines what should be the final result.

Blockchain provides a mechanism for creating tamper-evident records, while cryptographic hashing [7] provides a means of detecting changes. However, storing complete academic datasets directly on a blockchain is unsuitable because of storage and privacy considerations. Similarly, decentralized storage alone does not provide the complete verification workflow required by an academic institution.

This research therefore suggests a four-component system that differentiates between data ingestion, academic governance, generation of cryptographic proofs, and external verification.

The system has a life cycle of:

Component 3 → Component 2 → Component 1 → Component 4

Initial results are sent to and processed by component 3. Component 2 handles the academic review and controlled corrections prior to finalization. Component 1 creates cryptographic evidence from the finalized results by hashing with SHA-256, Merkle Trees, IPFS and blockchain anchoring. Component 4 uses that evidence to independently verify academic claims.

This research has made the following contributions:

- A full 4-component academic result lifecycle that includes ingestion, governance, proof creation and verification.
- A process for control and correction of academic results that clearly distinguishes between results that can be edited and finalized results.
- A cryptographic proof layer with a combination of SHA-256, Merkle Trees, IPFS and blockchain anchoring.

- A proof-index mechanism between candidate/module information and the corresponding cryptographic evidence.
- A privacy-preserving verification layer that uses Zero-Knowledge Proof techniques and cross-layer validation across blockchain anchors, IPFS data, record hashes, Merkle proofs, and the proof index.

2 RELATED WORKS

The potential of blockchain for enhancing the integrity and portability of educational credentials has been explored. Grech and Camilleri discussed the application of blockchain in education record and credential management [1]. Blockcerts demonstrated how digital credentials can be issued using the blockchain and how machine-verifiable certificates can be created [2]. Rooksby and Dimitrov explored the topic of university grading enabled by blockchain and drew attention to the connection of technical systems with institutional practices [3].

EduCTX proposed a framework to manage and port higher-education credits through a blockchain approach [9]. Higher-education registries have been explored in the context of blockchain and smart contracts in other studies [11]. While these approaches show promise of using distributed technologies for educational records, they do not necessarily form a comprehensive process from academic ingestion to BOE review, controlled corrections, finalization, cryptographic anchoring, or privacy-preserving verification.

IPFS offers content-addressed storage that can be used to store larger datasets off-chain [4]. This is helpful when the purpose of blockchain storage is mainly to store small integrity anchors. Another approach to proving statements without disclosing the underlying classified information is Zero-Knowledge Proof techniques [6].

The proposed system integrates these technologies while explicitly introducing an academic governance layer. The goal is thus not just to write academic outcomes onto a blockchain, but to provide a complete lifecycle where academic decisions are made prior to cryptographic finalization, allowing the finalized results to be independently verified.

3 RESEARCH GAP AND OBJECTIVES

3.1 Research Gap

Current solutions typically focus on specific parts of academic credential security, such as storing on the blockchain, authentication with digital credentials, decentralized storage, or privacy-conscious verification. The challenge of securing the complete lifecycle, however, cannot be addressed by these technologies alone:

How can an academic result be transformed from an initial submission into an officially finalized result and independently verified without relying solely on the issuing institution?

The proposed system addresses this gap by combining:

Ingestion → Governance → Cryptographic Integrity → Privacy-Preserving Verification

3.2 Research Objective

The main objective of this research is to develop an integrated academic result lifecycle that transforms initial result submissions into finalized results that can be cryptographically verified, while supporting controlled academic corrections and privacy-preserving third-party verification.

The specific objectives are:

- Implement controlled ingestions of results.
- Allow BOE review and correction.
- Maintain revision and audit information.
- Provide for special-concern processing prior to finalization; upon completion, generate cryptographic proofs.
- Store finalized proof data with IPFS.
- Anchor cryptographic proof information on the blockchain.
- Provide efficient proof lookup using candidate and module information.
- Allow independent academic reviews.
- Minimize data disclosure during verification.

4 SYSTEM ARCHITECTURE

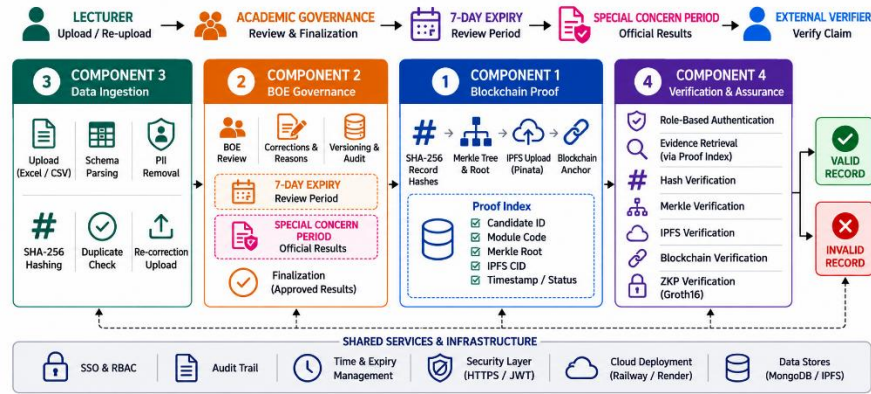


Fig. 1. End-to-end data flow and component pipeline of the BSAV system.

The components have separate responsibilities rather than duplicating functionality.

Component 3 — Data Ingestion

Component 3 is the starting point of academic result files. The lecturer's result sheets are processed and converted into the format required by the next stage. The component supports controlled submission and hand-off of the result information.

If corrections to the submitted file are needed during the permitted ingestion period, the corrected spreadsheet can be re-submitted through the controlled workflow.

Component 2 — BOE Governance

Component 2 is the academic review layer. Authorized BOE users can view candidate results, make controlled corrections, maintain version information, and record reasons for corrections.

The current architecture also separates the normal BOE review process from the special concern stage. Results are therefore subject to institutional control until they become final.

Component 1 then prepares the final approved result for cryptographic processing.

Component 1 — Blockchain Proof Layer

Component 1 gets the finalized records and their SHA-256 hashes from Component 2. It validates the hashes itself, generates Merkle Tree and Merkle Root, stores the finalized proof dataset on IPFS and anchors Merkle Root and IPFS CID via the ProofStorage smart contract.

Component 1 also maintains a MongoDB proof index that translates Candidate ID + Module Code into Merkle Root + IPFS CID.

Component 4 uses this index to look up the matching proof context. The implemented processing flow for Component 1 consists of the following six stages: receiving finalized records, hash validation, creating the Merkle Root, storing the finalized dataset on IPFS, proof anchoring, and proof lookup index.

Component 4 — Verification

Component 4 is the external verification layer. It does not generate or change academic results. Instead, it relies on the evidence produced by Component 1

Component 4 can request the relevant candidate/module proof context from Component 1 for a single academic claim, validate the blockchain anchor and IPFS dataset, regenerate the record hash, verify Merkle membership and perform the required Zero-Knowledge Proof verification before returning the verification decision.

5 COMPONENT 3 — DATA INGESTION

Component 3 is an access point for the controlled submissions of academic results.

The ingestion process executes the following generic operations:

- Submission from Lecturer: The React/Vite portal where lecturers can drag & drop their Excel/CSV sheets.
- Processing spreadsheets and extraction of results: The SheetJS parser dynamically identifying the Candidate ID and harvesting the assessment columns irrespective of the Excel layout.
- Standardization and privacy filtering: Sorting columns alphabetically for deterministic hashing, and stripping of PII (names, emails, phone numbers) for PDPA compliance.
- Duplicate and integrity checking: Generating the SHA-256 payload hash to instantly reject duplicate files and prevent database bloat.
- Provenance handling: Anchoring the verified data into the MongoDB Atlas private ledger as an immutable, chained block (Block #0, Block #1, etc.).
- Controlled resubmission: The intelligent deep-patching engine that applies candidate-specific grade appeals without erasing historical class records.
- Time-gated submission: The policy engine uses timestamps to automatically enforce the Standard Window, BOE Lock, and Special Concerns phases.
- Controlled handoff to Component 2: The background watcher and direct override API that push the clean, validated data to the Board of Examiners

The component does not give final cryptographic anchoring but rather gives consistent result information for the academic governance layer.

The architecture also allows re-submission in case an upload needs to be corrected during the submission process.

6 COMPONENT 2 — CONTROLLED BOE GOVERNANCE

6.1 BOE Review and Correction

Component 2 provides a controlled interface for authorized BOE members to review academic results.

The implemented functionality includes:

- authentication.
- module-based review.
- candidate search.
- controlled result revision.
- automatic grade calculation.
- correction reasons.
- revision history.
- version tracking.

Previous versions are retained through revision history to provide traceability of controlled changes. The changes are recorded in the revision process.

6.2 Review and Special-Concern Periods

Currently, the regular BOE review process is distinguished from the special-concern stage that follows.

This stage provides a controlled period during which identified concerns can be reviewed and resolved before finalization.

The important design principle is:

Review → Concerns → Finalization → Cryptographic Anchoring

This prevents Component 1 from anchoring results where academic change is still expected.

6.3 Finalization

Once the relevant academic review and concern processes have been completed, the approved result is finalized.

Component 2 generates the SHA-256 hash for each finalized student record and transfers the finalized records and hashes to Component 1.

7 COMPONENT 1 — CRYPTOGRAPHIC PROOF LAYER

7.1 SHA-256 Validation

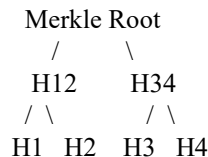
Component 1 receives finalized records together with their hashes. In the current implementation, the canonical record representation follows the format.

candidate | moduleCode | marks | grade | version.

Component 1 independently reconstructs this representation and calculates the SHA-256 hash [7] before accepting the received hash. A mismatch indicates that the received record does not correspond to the expected cryptographic representation.

7.2 Merkle Tree Construction

The validated record hashes are used as leaves of a binary Merkle Tree [5].



The Merkle Root provides a compact representation of the complete finalized batch. If one committed record changes, its hash changes and the resulting Merkle Root also changes. The Merkle structure additionally supports membership proofs for individual records. The Merkle Root is subsequently used as the batch-level cryptographic commitment that is anchored on the blockchain.

7.3 IPFS Storage

After generating the Merkle Root, Component 1 creates the finalized proof dataset and uploads it to IPFS [4] through Pinata. The IPFS CID identifies the stored content.

The system therefore separates:

Large proof dataset → IPFS

from:

Compact cryptographic anchor → Blockchain

The current implementation uses public IPFS as a prototype storage mechanism. Therefore, IPFS should not be presented as the privacy boundary of the system. Production deployment would require appropriate encryption or controlled/private storage for sensitive academic information.

7.4 Blockchain Anchoring

Component 1 uses the ProofStorage Solidity smart contract [8] to anchor the cryptographic proof.

The contract records:

- Merkle Root.
- IPFS CID.
- timestamp.
- uploader address.

The contract also prevents an already anchored Merkle Root from being anchored again. This means that the blockchain does not contain the complete academic dataset. Instead, it provides the tamper-evident cryptographic reference against which the external proof data can later be checked.

7.5 Hardhat EVM Environment

The current research prototype uses a Hardhat local EVM [8] environment.

Hardhat provides the development and testing environment, while the EVM executes the Solidity smart contract. This should be described as a prototype blockchain environment, not as a production public blockchain. A production implementation would migrate the contract to a persistent EVM-compatible network or an appropriate institutional/consortium blockchain.

7.6 MongoDB Proof Index

MongoDB is used separately from IPFS. Its purpose is to provide fast application-level lookup.

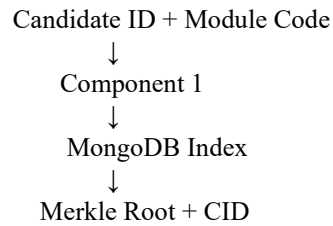
The index stores information such as:

- candidate ID.
- module code.
- Merkle Root.
- IPFS CID.
- anchoring time.

Therefore: IPFS stores the proof dataset, MongoDB finds the proof, and block-chain anchors the proof. This separation avoids treating MongoDB as the ultimate integrity source.

7. 7 Component 4 Integration

Component 4 can request:

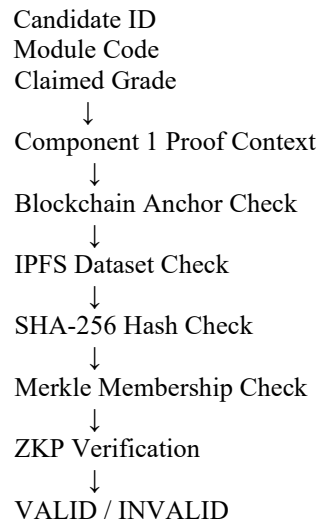


Component 4 then uses the returned proof context to access the required block-chain and IPFS evidence.

The documented integration includes GET /proof/record/{candidateId}/{module-Code} for proof-context discovery, followed by Component 1 endpoints for block-chain proof, finalized proof data, and Merkle proof retrieval.

8 COMPONENT 4 — PRIVACY-PRESERVING

In this step, Component 4 uses Component 1 as the source of the finalized cryptographic evidence. For a single academic claim, the implemented verification process is shown below:



The workflow binds the proof to the candidate, module, claimed grade, and Merkle Root. This helps prevent the same proof from being used for a different verification claim. Component 4 also provides additional security checks using Groth16-based ZKP verification [10] and Certora-based formal verification of the smart contract.

9 SECURITY AND PRIVACY MODEL

Table 1. Security Layers' Purposes

Layer	Purpose
SHA-256	Record integrity
Merkle Tree	Batch integrity and membership
IPFS	Finalized proof dataset storage
Blockchain	Trusted cryptographic anchor
ZKP	Privacy-preserving verification
BOE workflow	Academic governance
Audit/versioning	Traceability

The system does not rely on one technology for all security requirements.

The resulting trust chain is:

Provenance → Governance → Integrity → Storage → Blockchain Anchor → Privacy-Preserving Verification

10 EVALUATIONS

The current prototype was evaluated using component-level and integration testing. The evaluation focused on checking the main functions of the system, including result transfer between components, hash validation, Merkle Root generation, IPFS storage and retrieval, blockchain anchoring, proof lookup, and integrity verification. The tests were carried out using the current implementation to check whether the complete verification process works correctly.

10.1 Functional Evaluation

The current prototype was tested using component-level and integration testing. The main tests include:

- Finalized result transfer from Component 2.
- SHA-256 Hash validation.
- Merkle Root generation.
- IPFS upload and retrieval.
- Blockchain proof anchoring.
- Prevention of duplicate anchors.
- Proof index creation.
- Latest proof retrieval.
- Historical proof retrieval.
- Data integrity verification.
- Component 4 proof lookup.

The current system successfully generates stores and retrieves proof information using IPFS and the blockchain proof workflow.

10.2 Integrity Evaluation

The integrity check uses several layers to make sure the data is correct:

- Blockchain proof.
- IPFS dataset.
- SHA-256 hashes.
- Merkle Root.
- MongoDB proof index.

For a valid finalized dataset, all these checks should pass. If a stored record is changed, its SHA-256 hash will also change. This will affect the Merkle Root and allow the system to detect the change.

10.3 Prototype Limitations

The current system has two main limitations. The system uses a local Hardhat blockchain, so the blockchain data is not permanent. Restarting the local network can reset the blockchain state. The system currently uses public IPFS. Therefore, better privacy and security are needed before storing real academic data.

These limitations show the areas that need to be improved before using the system in a real-world environment.

11 DISCUSSIONS

The main contribution of our system is separating academic result management from cryptographic verification.

- Academic results are reviewed and finalized before creating the proof.
- Component 1 checks the hashes before creating the Merkle Root.
- IPFS stores proof data.
- Blockchain anchors Merkle Root and IPFS CID.
- MongoDB helps to quickly find the correct proof using academic details.

Each technology has a separate role in the system.

This approach reduces the need to depend on a single technology and provides a more organized and secure verification process.

12 NOVELTIES

The main novelty of our system is combining academic result management and secure verification into one complete process. Academic results are reviewed and finalized before creating the proof.

The system includes:

- Controlled academic result review.
- Version-based corrections and special case handling.
- SHA-256 Hash validation.
- Merkle-based batch verification.
- IPFS for proof storage.

- Blockchain anchoring of the Merkle Root and IPFS CID.
- Proof-based result verification.
- Privacy-preserving verification.

Each technology has a separate role in the system.

This makes the system more than just a blockchain-based academic result system because it combines academic management, data integrity, and secure verification in one process.

13 LIMITATIONS AND FUTURE WORK

The current prototype can be improved in the future by:

- Academic results are reviewed and finalized before creating the proof.
- Moving from local Hardhat to a permanent blockchain network.
- Improving privacy for academic data storage.
- Testing with larger and different datasets.
- Expanding formal verification for smart contract security.
- Improving secure and efficient IPFS proof retrieval.
- Testing the system with anonymized university data.
- Improving verification of historical proofs from different finalized batches.
- Adding stronger access control, key management, monitoring, and governance.

14 CONCLUSION

This paper presented a four-component architecture for secure and independently verifiable academic results. The system starts with controlled results in ingestion in Component 3. Component 2 handles academic review, corrections, special concerns, and finalization. Component 1 validates the finalized record hashes, creates the Merkle Tree, stores the proof dataset on IPFS, anchors the Merkle Root and IPFS CID on the blockchain, and maintains the MongoDB proof index. Component 4 uses this evidence for independent and privacy-preserving verification.

The main idea of the system is not to depend only on blockchain for security. Instead, trust is built step by step:

Provenance → Governance → Integrity → Blockchain Evidence → Privacy → Verification

The current prototype shows how these layers can work together. However, future improvements are needed before real-world deployment, especially a permanent blockchain network and stronger privacy controls for sensitive academic data.

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